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Explain the term "Modeling" in context of 3D coordinate system.

Answer ([Please click here to Add Answer](#))



Looking at the Wings 3D workspace, you can see the red, blue, and green lines that represent the X,Y, and Z axes.

In Wings 3D, Y is up and down, X is side to side, and Z is front to back. These three axes are commonly used to define the active axis for moving, scaling, or rotating a model. Often in Wings 3D they will be referred to as Standard axes.

Note: Other modelers may swap the Y and Z axis, so when importing a model into Wings 3D it may appear to be the wrong way around.

Points

A point is any location in 3D space. Points can be used as references, such as the center around which a model will be rotated. A point may be defined by the intersection of two vectors. Points can also be used to define the location of a plane.

We want to scale an object two times of the existing size on X-axis and Y-axis. Write the scaling matrix for this transformation.

Answer ([Please click here to Add Answer](#))



What would you understand by the following matrix in context of clipping plane?

$$\begin{bmatrix} w & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & Q & 1 \\ 0 & 0 & -QZ_s & 0 \end{bmatrix}$$

Answer ([Please click here to Add Answer](#))



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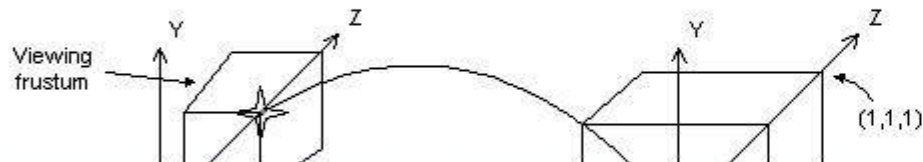
How can we perform the Shearing transformation? Also discuss its two types.

Answer ([Please click here to Add Answer](#))



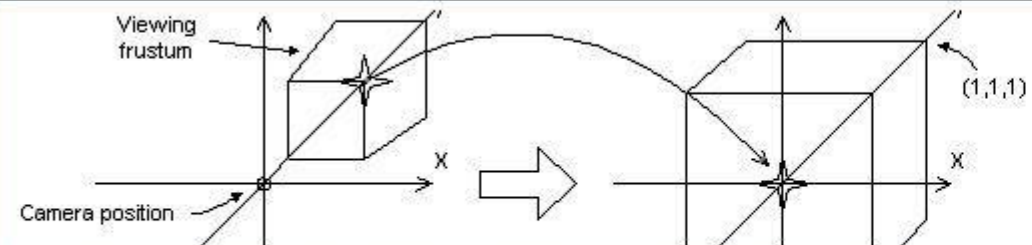
Transformation means changing some graphics into something else by applying rules. We can have various types of transformations such as translation, scaling up or down, rotation, shearing, etc. When a transformation takes place on a 2D plane, it is called 2D transformation.

What following figure illustrates about the perspective transformation being performed

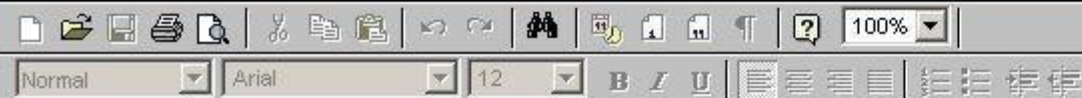


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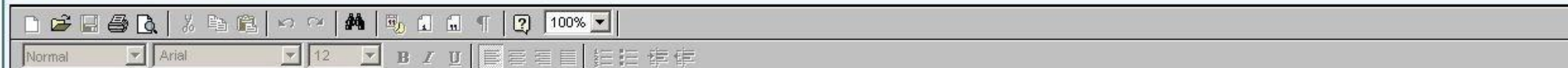


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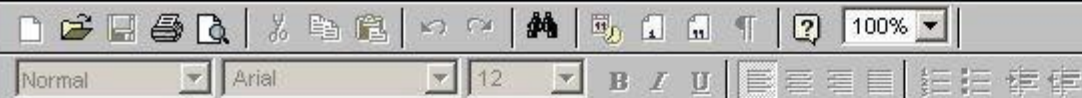


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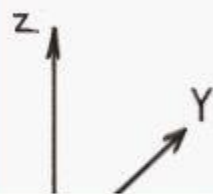


An object is scaled and rotated about its centroid coordinates (x_c, y_c) and then translated. You are required to find the values for the elements of the composite transformation matrix.

Answer ([Please click here to Add Answer](#))

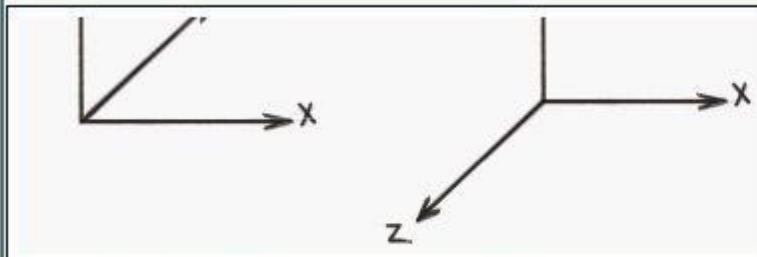


Write down which type of professionals uses y-up versus z-up coordinate system?



Answer ([Please click here to Add Answer](#))





Answer ([Please click here to Add Answer](#))



Explain the term "Modeling" in context of 3D coordinate system.

Answer ([Please click here to Add Answer](#))



In which situation intersection between a Line and Plane occurs?

Answer ([Please click here to Add Answer](#))



How can we express the rotation of an object by using Homogeneous Coordinates in matrices form?

Answer ([Please click here to Add Answer](#))



When we discuss 3D transformations, explain the concept of translation with the help of example.

Answer ([Please click here to Add Answer](#))



Apply the following transformations on the point $P(x,y) = (4,5)$.

1. Translate using $T_x = 3$ and $T_y = 2$
2. Scale using $S_x = 2$ and $S_y = 1$

Answer ([Please click here to Add Answer](#))



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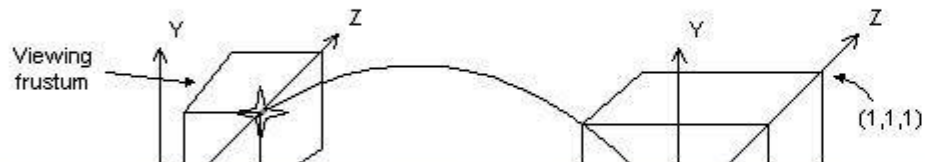
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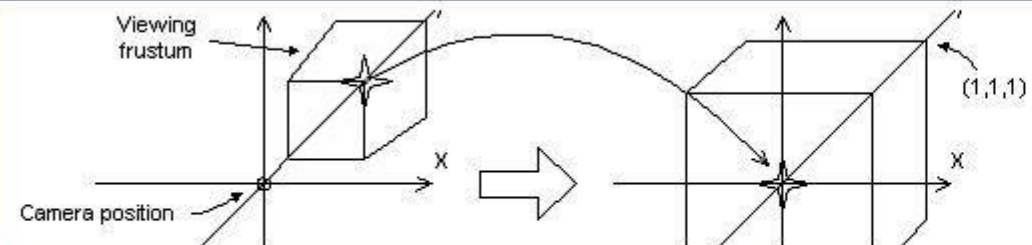
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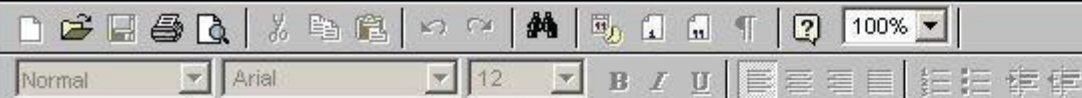


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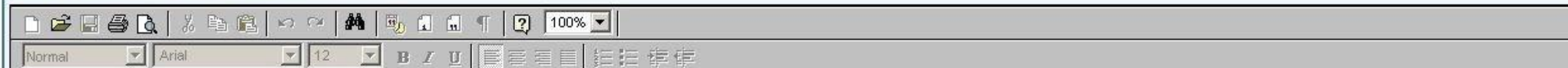


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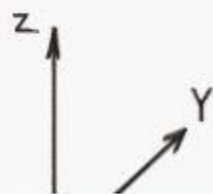


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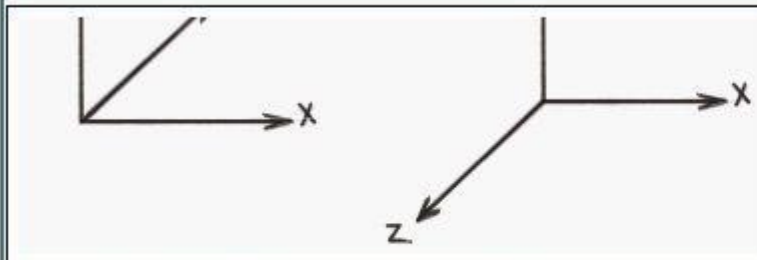


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